

Customer Churn Prediction — Business Insights

Project Objective

The objective of this project was to analyze customer behavior and predict churn using machine learning. Customer churn refers to customers leaving a service provider, which directly impacts revenue and business growth.

Key Dataset Summary

- Total Customers Analyzed: 7,032
- Customers Churned: 1,869
- Churn Rate: 26.58%
- Average Monthly Charges: 64.8
- Average Customer Tenure: 32.42 months

Key Insights

1. High Overall Churn Rate

Approximately 26.58% of customers left the company, indicating significant customer retention challenges.

2. Contract Type Strongly Impacts Churn

Customers with month-to-month contracts showed the highest churn probability, while long-term contract customers were more likely to stay.

3. Fiber Optic Customers Show Higher Churn

Customers using fiber optic internet had higher churn rates compared to DSL and no-internet customers, suggesting possible pricing or service quality issues.

4. Payment Method Matters

Customers using electronic check showed the highest churn behavior. Automatic payment users demonstrated better retention.

5. Monthly Charges Influence Churn

Higher monthly charges were strongly associated with increased churn probability.

6. Short Tenure Customers Are At Risk

New customers with low tenure were more likely to churn than long-term customers.

Machine Learning Results

A Logistic Regression model was trained to predict customer churn using customer demographics, service usage, and billing data.

Model Evaluation:

- Algorithm: Logistic Regression

- Model Accuracy: ~80%
- Evaluation Metrics Used:
 - Accuracy
 - Precision
 - Recall
 - F1 Score
 - Confusion Matrix

Business Recommendations

- Offer discounts to month-to-month customers to encourage long-term contracts.
- Investigate service quality issues for fiber optic customers.
- Promote auto-payment methods for better customer retention.
- Introduce early retention campaigns for new customers.
- Monitor high monthly charge customers with personalized offers.

Conclusion

The project successfully identified major churn drivers and built a predictive model to help businesses proactively reduce customer loss and improve retention strategies.